Arpit Jain

Platform Engineer | Hybrid Cloud & AI Infrastructure | Kubernetes | Terraform | Go

Bengaluru, Karnataka, India | +91 9672590654 | arpit2k01jain@gmail.com | LinkedIn

Professional Summary

DevOps and Platform Engineer with more than 3 years of experience working on production infrastructure across AWS, Kubernetes, OpenShift, and bare-metal environments. Experienced in infrastructure automation using Terraform, GitOps with ArgoCD, CI/CD, and Go-based tooling. Worked on high-performance networking and workloads using DPDK, SR-IOV, and Cilium, distributed storage with Ceph and Mayastor, and multi-cluster management with OpenShift ACM. Hands-on experience in infrastructure security and compliance using OpenSCAP, OVAL, Kubescape, kube-bench, and Kyverno, including improving CIS Level 2 server compliance from 35% to 85+%.

Education

Indian Institute of Technology (IIT), Palakkad 2019 – 2023
B.Tech in Civil Engineering *CGPA: 8.33/10*

Technical Skills

Cloud & Infrastructure: AWS (EKS, EC2, Lambda, ECR, IAM, VPC, S3, SNS, SQS, ACM PCA), Kubernetes, OpenShift/OCP, OpenShift ACM, KubeVirt

Infrastructure as Code & Automation: Terraform, Ansible, Helm, GitHub Actions, Tekton, ArgoCD

Networking & Storage: SR-IOV, Cilium, BGP, MetalLB, Traefik, Ceph, OpenEBS Mayastor, Load Balancing, Service Mesh

Security & Compliance: IAM, RBAC, CIS Level 2, OpenSCAP, OVAL, Trivy, Kubescape, kube-bench, Kyverno, Pod Security Standards, Wazuh

Observability: Prometheus, Grafana, Victoria Metrics, OpenTelemetry, CloudWatch, Loki

Programming & Systems: Go, Python, Bash/Shell, Linux, Linux Kernel Tuning, OS Patching & Upgrades

Professional Experience

Nava April 2026 – Present
Platform Engineer / DevOps Engineer *Bengaluru, India*

- **Cloud Infrastructure & IaC:** Designed and implemented production-grade, multi-account AWS infrastructure across Dev, Staging, Pre-Production, and Production using Terraform, covering EKS, EC2, Lambda, ECR, ACM PCA, SNS, SQS, IAM, VPC, and S3. Built reusable modules, secure cross-account access using IAM Roles Anywhere and Trust Anchors, and GitHub Actions workflows for automated plan, approval-gated apply, and controlled destroy operations.
- **Platform Engineering & Kubernetes:** Architected and operated production-grade vanilla Kubernetes clusters for high-performance workloads, integrating Cilium/eBPF, Ceph, Mayastor, KubeVirt, SR-IOV, and MetalLB for advanced networking, distributed storage, virtualization, and workload connectivity.
- **DNS Platform Engineering:** Engineered a scalable Go-based DNS management platform integrating PowerDNS and Akamai DNS, automating DNS operations while providing request-level metering and billing visibility for enterprise-scale zones.
- **GitOps & Release Automation:** Implemented GitOps-based deployment workflows using ArgoCD and GitHub Actions, automating application delivery and platform changes with controlled rollout strategies including blue/green and canary deployments.
- **Security & Compliance:** Led infrastructure security hardening using OpenSCAP, OVAL, Kyverno, Kubescape, and kube-bench, improving CIS Level 2 server compliance from 51% to 85+% and Kubernetes security compliance from 40% to 80%.
- **Observability & Reliability:** Developed custom node security audit tooling to track vulnerability remediation and compliance deltas, and implemented full-stack platform observability using the OpenTelemetry ecosystem for metrics, logs, and traces.

Jio Platforms Limited October 2023 – April 2026
SRE/DevOps Engineer *Mumbai, India*

- **Private Cloud & Air-Gapped Infrastructure:** Operated production-grade OpenShift (OCP) clusters in highly restricted, disconnected environments, managing private container registries, OIDC authentication, custom Operator Hub workflows, and platform-level configurations.
- **High-Performance Networking & Workloads:** Optimized CPU- and network-intensive DPDK-based telco applications using CPU pinning, topology-aware scheduling, and SR-IOV/OVS networking to support low-latency, high-throughput workloads.
- **Cluster Provisioning & Automation:** Engineered Tekton-based CI/CD pipelines to automate OpenShift cluster provisioning and infrastructure validation, including custom Ignition-based ISO images for automated server connectivity and pre-deployment checks.
- **Multi-Cluster Management:** Implemented centralized cluster lifecycle and operational management using OpenShift Advanced Cluster Management (ACM), enabling consistent configuration and management across multiple on-premises OpenShift environments.

- **Production Reliability & Troubleshooting:** Diagnosed and resolved complex production issues across application, Kubernetes/OpenShift, Linux kernel, networking, and distributed storage layers, reducing service disruption and improving platform reliability.

Key Projects

Production-Grade Multi-Account AWS Infrastructure | *Terraform, AWS, GitHub Actions*

- Designed and implemented a production-grade multi-account AWS infrastructure across Dev, Staging, Pre-Production, and Production environments using Terraform, with GitHub as the single source of truth.
- Built reusable, modular Terraform components for EKS, EC2, Lambda, ECR, ACM PCA, SNS, SQS, IAM, and other AWS services to standardize infrastructure provisioning across environments.
- Implemented secure cross-account access using AWS IAM Roles Anywhere, Trust Anchors, and role-based authentication, reducing dependency on long-lived credentials.
- Developed GitHub Actions CI/CD workflows for automated Terraform formatting, validation, plan generation, approval-gated apply, and controlled destroy operations.

Production-Grade Telco Cloud Platform on Bare Metal | *OpenShift, ACM, Kubernetes, CI/CD*

- **Telco Cloud Platform:** Designed and deployed a production-grade, air-gapped telco cloud platform on bare-metal infrastructure from the ground up, including private container registry, private DNS, OpenShift ACM, Operator Hub, organizational certificates, and platform services.
- **Large-Scale Cluster Deployment:** Built and validated a single-node OpenShift ACM management cluster and used it to provision and manage production telco cloud clusters across 500+ bare-metal nodes, enabling centralized cluster lifecycle and application management.
- **Network & Platform Integration:** Automated deployment and configuration of SR-IOV, Network Attachment Definitions (NADs), MetalLB, telemetry, and cluster networking components required for high-performance telco workloads.
- **CI/CD & Automation:** Engineered an end-to-end CI/CD pipeline to pre-validate network connectivity, provision clusters, perform post-deployment configuration, deploy operators and applications, and execute platform validation automatically.
- **End-to-End Platform Engineering:** Automated the complete lifecycle from bare-metal readiness and cluster provisioning to post-deployment configuration and application onboarding, significantly reducing manual intervention and enabling repeatable production deployments.

Infrastructure Security & Compliance Automation Platform | *OpenSCAP, OVAL, Trivy, Kubescape, kube-bench, Kyverno*

- **Node Security Audit Platform:** Designed and developed a multi-node security audit platform with a TUI-based interface to orchestrate concurrent deep scans across infrastructure nodes using OpenSCAP, OVAL, and custom node audit tooling.
- **Vulnerability Assessment:** Automated collection and analysis of CVE findings, including severity, vulnerability status, affected packages, installed versions, and available versions containing fixes, with consolidated reports for infrastructure security assessment.
- **Security Remediation:** Used the platform to identify, track, and remediate infrastructure vulnerabilities and configuration gaps, improving the CIS Server Level 2 compliance score to 85+% across the assessed infrastructure.
- **Kubernetes Security & Compliance:** Implemented Kubernetes security assessments and policy enforcement using Kubescape, kube-bench, and Kyverno, covering CIS benchmarks, cluster configuration, workload security, and compliance policies.
- **DevSecOps & CI/CD:** Integrated Trivy into GitHub Actions CI/CD workflows to perform container and dependency vulnerability scanning before container image build and publication, enforcing security checks early in the software delivery lifecycle.

Automated Certificate Lifecycle & Monitoring | *Kubernetes, cert-manager, Prometheus, Grafana*

- **Certificate Lifecycle Automation:** Designed and implemented an end-to-end certificate lifecycle management solution covering Kubernetes certificates, service certificates, database certificates, and other platform TLS certificates, automating certificate provisioning, renewal, and rotation.
- **Expiry Monitoring & Alerting:** Implemented centralized certificate expiry monitoring with proactive alerts before certificate expiration, enabling timely remediation and preventing certificate-related production outages.
- **Automated Renewal:** Integrated automated renewal and rotation workflows using cert-manager for Kubernetes workloads, ensuring certificates were renewed and deployed without manual intervention or service downtime.